MPC Approach to Actuator Fault Accommodation Through $\mathcal{H}_2$-Norm Minimization, with Application to Cascading CSTRs

Frank Vilimek
Department of Chemical Engineering
University of California, Davis
Davis, CA, United States
fjvilimek@ucdavis.edu

Jaden Oh
Department of Mechanical and Aerospace Engineering
University of California, Davis
Davis, CA, United States
jmjoh@ucdavis.edu

***Abstract*—We explore a method of accommodating multiplicative actuator faults through a modified fault parameter that minimizes the $\mathcal{H}_2$-norm, applied to a cascading CSTR system with model-predictive control. We find that fault accommodation is often the highest-performing when the accommodated fault parameter is equal to the estimated fault, indicating that minimal accommodation-related calculations are required once the fault is estimated; fault accommodation was explored with a deterministic system and a system with stochastic measurement noise to probe its robustness. The fault estimation and accommodation method is also shown to work well with multiple faults, each of varying magnitude. We explore the effects of changing the size of the set of input constraints and how it alters the shape of the $\mathcal{H}_2$-norm surface, where we see a clear dependence of performance on maximum input constraints; we reach the literature-backed conclusion that unconstrained MPC produces a state trajectory that is both optimal in terms of the quadratic state-input cost and the $\mathcal{H}_2$-norm. Lastly, MPC was shown to produce state and input trajectories that were lower than those produce by LQR, further validating the argument that MPC is preferable over other control methods when system performance is of importance.**

***Index Terms*—model-predictive control, fault-tolerant control, fault estimation, fault accommodation, H2**

I. Introduction

Since their adoption into industrial processes, control systems have played a crucial role in stabilizing and controlling the states of dynamic systems. The abstraction of a physical process into a mathematical framework has allowed control systems to be rigorously studied and the properties and behavior of such systems has been an active area of research since its conception. Though the use of mathematics is powerful in designing controllers, the extent of its utility goes only as far as the extent and accuracy of the model used to describe them.

Real-world systems are far from ideal, and events such as plant, sensor, or actuator faults must be accounted for in order to have a control system robust to such disturbances. In the context of fault-susceptible systems, a fault-tolerant control (FTC) method must be used in order to mathematically model the fault, estimate its magnitude, and accommodate it through adjusting the calculation of the control input. As we need to be able to model a fault-susceptible control system, the choice of control method is an important consideration.

Among the many control methods used across major industries, proportional-integral-derivative (PID) control remains the one with the most widespread application [1]. However, though PID controllers are simple to design and implement, they are inherently only reactive to process disturbances and are unable to provide a control input based in optimality (though work has been done to incorporate optimization in terms of performance [2]). Their modeling is also unable to take state and input constraints into account, making controller saturation and any resulting instability key downsides.

Switching to a model-based control method such as full-state feedback addresses many of the issues of PID control: model-based control methods can predict how a system will react to inputs or disturbances and are thus better equipped to model faults and their impacts on system behavior. Model-based control methods are also able to deal with multiple states, inputs, and faults, all of which can be simultaneously calculated. However, model-based control methods like full-state feedback are not inherently able to calculate a control input that is optimal, and control methods that are optimal such as the linear-quadratic regulator (LQR) are not able to take state and input constraints into account, leaving room for improvement.

Model-predictive control (MPC) is a control method that fills this gap of constrained optimization. Devised specifically as a model-based, constrained optimal control method, MPC is perhaps the most well-equipped to model, estimate, and accommodate faults in real-world systems where state and input constraints pose real challenges. Much progress has been made with MPC in the domain of FTC, where utilizing optimization-based fault estimation and accommodation methods as duals to MPC have proven fruitful. It is in this same vein that we approach a method of fault accommodation previously only applied to systems with non-optimal model-based control and investigate the extent to which it applies. Introduced by Allen and El-Farra [3], the method requires an estimate of the fault to be made, after which a modified fault parameter is calculated to force the controller to believe a fault of a different magnitude has occurred, speeding up the controller's response to reduce the settling time. Though still exploratory, this work aims to investigate a small yet potentially useful arm of the FTC space.

II. Background

The body of work surrounding fault-tolerant control is quite rich with a variety of formulations for estimating and accommodating different types of faults.

Napasindayao and El-Farra's work on applying gain scheduling methods for FTC focused heavily on ensuring exponential stability of the controlled system, providing a strong foundation for studying multiplicative faults and FTC design around stability requirements. The main drawback of this work is that it uses full-state feedback and a linear control law, where the poles of the transfer function were chosen only to guarantee stability without optimality or constraints [4].

Allen and El-Farra's work with estimating a fault parameter, then recalculating it to improve controller performance by minimizing the $\mathcal{H}_2$-norm, is a novel addition to the body of fault accommodation strategies and serves as the inspiration for this project. The work also considers times when sensor noise could potentially impact fault parameter estimation, and provides a method for avoiding any incurred instability. However, similar to Napasindayao and El-Farra's work, the use of full-state feedback means that this FTC strategy lacks optimal constrained control [3].

Sun et al.'s work on FTC design using MPC with input constraints and actuator faults is able to merge the constrained optimization of MPC with fault detection and accommodation, however the main drawback in their work with regards to fault estimation is that it is limited to faults that decrease the range of control inputs [5]. Their method also does not explicitly calculate a fault parameter, which could be useful in assisting the type of fault at hand for adjusting repair plans.

Xu et al. combines a tube-based MPC with set-theoretic fault detection and isolation (FDI) in the presence of actuator faults [6]. The fault detection is handled passively by invariant sets, while the fault isolation is performed actively. After a fault has been detected, using the MPC's innate constraint-handling capability as the accommodation method itself, it adjusts the input-constraint set of the MPC controller. In a companion work by the same authors, they apply an analogous robust-MPC and set-based estimation framework to sensor faults. The fault isolation is gained by exploiting MPC to decouple the effect of each sensor fault on a corresponding output channel [7]. While these works successfully integrate optimal constrained control with FDI, the fault accommodation step is a switching or constraint-tightening procedure, rather than a recalculation of an estimated multiplicative fault parameter, which again may prove useful for repair actions.

Bououden et al. proposes a robust fault-tolerant MPC for discrete-time linear systems subject to simultaneous sensor and actuator faults, disturbances, and input constraints [8]. They first design a virtual observer to improve fault estimation accuracy, then construct a real, implementable observer on top of it, and use the resulting state and fault estimates inside a linear-matrix inequality (LMI) for the MPC. This guarantees robust stability and bounded-disturbance rejection. Their work handles both fault classes within a single optimization-based controller, but it does not use the fault estimate to actively re-tune the controller's performance objective, which our project aims to achieve.

III. Project Description and Methodology

A. The Dynamic Model: Cascading CSTRs

Continuously-stirred tank reactors (CSTRs) are widely used across the chemical- and biochemical-related industries for their overall simplicity in design and ease of temperature control [9]. CSTRs most commonly are equipped with impellers that stir its contents such that everything—reactants, products, inerts, etc.—are combined in a homogeneous mixture. This allows us to mathematically model them as having no spatial dependence within the reactor, simplifying their model to be only dependent on time. This simplification, along with their widespread use across multiple industries, makes CSTRs the benchmark for studying control systems in the context of chemical processing.

The model we choose to study is a set of cascading CSTRs. The first CSTR contains the liquid-phase reaction $A+B \rightarrow C$, along with a heating jacket to control the internal temperature. The second CSTR is fed the outlet of the first CSTR along with an inlet stream of reactants; this CSTR contains the liquid-phase reactions $A + B \rightarrow C$ and $C + D \rightarrow E$, along with its own heating jacket. A process flow diagram of this system is observed in figure (1).

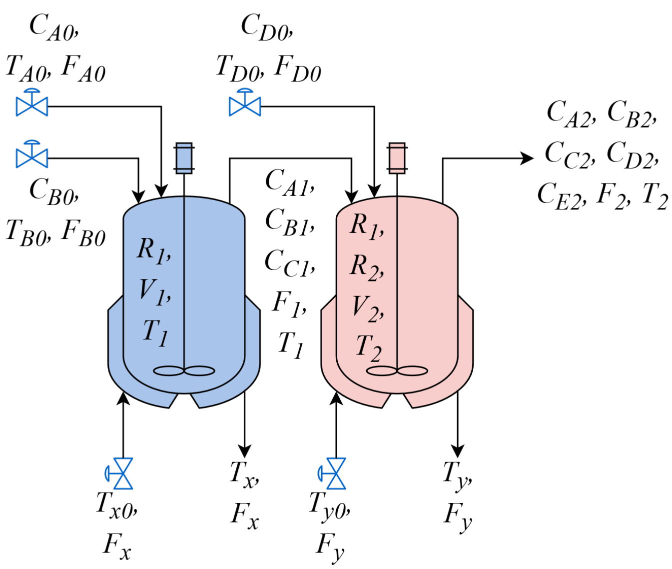


Fig. 1: The system used in this study is a set of cascading CSTRs, each with multiple inlets for reactants in the main tanks and heat transfer fluid in the heating jackets.

We are interested in controlling a total of twelve state variables: the concentrations of species A, B, and C in CSTR 1 (C_{A1}, C_{B1}, and C_{C1}, respectively); the concentrations of species A, B, C, D, and E in CSTR 2 (C_{A2}, C_{B2}, C_{C2}, C_{D2}, and C_{E2}, respectively); and the temperature of CSTR 1 and 2, and the temperature of the heating jackets around CSTR 1 and 2 (T_1, T_2, T_x, and T_y, respectively). Each of these state variables obeys system dynamics described through a set of twelve coupled nonlinear differential equations:

$$\frac{dC_{A1}}{dt} = \left(\frac{F_{A0}C_{A0} - F_1C_{A1}}{V_1}\right) - R_1^{(1)} \tag{1}$$

$$\frac{dC_{B1}}{dt} = \left(\frac{F_{B0}C_{B0} - F_1C_{B1}}{V_1}\right) - R_1^{(1)} \tag{2}$$

$$\frac{dC_{C1}}{dt} = -\left(\frac{F_1C_{C1}}{V_1}\right) + R_1^{(1)} \tag{3}$$

$$\frac{dC_{A2}}{dt} = \left(\frac{F_1C_{A1} - F_2C_{A2}}{V_2}\right) - R_1^{(2)} \tag{4}$$

$$\frac{dC_{B2}}{dt} = \left(\frac{F_1C_{B1} - F_2C_{B2}}{V_2}\right) - R_1^{(2)} \tag{5}$$

$$\frac{dC_{C2}}{dt} = \left(\frac{F_1C_{C1} - F_2C_{C2}}{V_2}\right) + R_1^{(2)} - R_2^{(2)} \tag{6}$$

$$\frac{dC_{D2}}{dt} = \left(\frac{F_1C_{D0} - F_2C_{D2}}{V_2}\right) - R_2^{(2)} \tag{7}$$

$$\frac{dC_{E2}}{dt} = -\left(\frac{F_2C_{E2}}{V_2}\right) + R_2^{(2)} \tag{8}$$

$$\frac{dT_1}{dt} = \left(\frac{F_{A0}T_{A0} + F_{B0}T_{B0} - F_1T_1}{V_1}\right) + \frac{(-\Delta H_1)}{\rho_r C_{pr}}R_1^{(1)} - \frac{U_xA_x}{\rho_r C_{pr} V_x}(T_1 - T_x) \tag{9}$$

$$\frac{dT_2}{dt} = \left(\frac{F_1T_1 + F_{D0}T_{D0} - F_2T_2}{V_2}\right) + \frac{(-\Delta H_1)}{\rho_r C_{pr}}R_1^{(2)} + \frac{(-\Delta H_2)}{\rho_r C_{pr}}R_2^{(2)} - \frac{U_yA_y}{\rho_r C_{pr} V_y}(T_2 - T_y) \tag{10}$$

$$\frac{dT_x}{dt} = \frac{F_x}{V_x}(T_{x0} - T_x) + \frac{U_xA_x}{\rho_c C_{pc} V_x}(T_1 - T_x) \tag{11}$$

$$\frac{dT_y}{dt} = \frac{F_y}{V_y}(T_{y0} - T_y) + \frac{U_yA_y}{\rho_c C_{pc} V_y}(T_2 - T_y) \tag{12}$$

where $F_1 = F_{A0} + F_{B0}$; $F_2 = F_1 + F_{D0}$; and $R_1^{(i)} = A_1e^{-E_1/R_gT_i}C_{Ai}C_{Bi}$ and $R_2^{(i)} = A_2e^{-E_2/R_gT_i}C_{Ci}C_{Di}$, i indicating in which CSTR the reaction is occurring. All constants, their description, and the values chosen for simulation are tabulated in the appendix. This model assumes:

- the tank contents are well-mixed,
- the heat transfer fluid in the heating jackets is well-mixed,
- the chemical reaction rate constant is Arrhenius in behavior,
- the reactants and products are dilute and their impact on fluid density and heat capacity is negligible,
- and fluid density and heat capacity for both the mixture and heat transfer fluid are constant with temperature.

The manipulated variables are the inlet flow rates of each reactant and heat transfer fluid (F_{A0}, F_{B0}, F_{D0}, F_x, and

F_y). The model was linearized about the steady-state control input $u_{eq} = [F_{A0}^{sp}\ F_{B0}^{sp}\ F_{D0}^{sp}\ F_x^{sp}\ F_y^{sp}]^\top$ to which the results can be found in the Appendix.

B. Discrete State-Space Model with Fault Modeling

In this study we focus on actuator faults that are multiplicative to the control input, where we borrow the formulation from Allen and El-Farra's work [3]. The discrete-time LTI system is described by the state-space model,

$$x_{k+1} = Ax_k + B\theta u_k \tag{13}$$

$$\hat{x}_{k+1} = \hat{A}\hat{x}_k + \hat{B}\hat{\theta} u_k \tag{14}$$

$$y_k = Cx_k, \tag{15}$$

where $x \in \mathbb{R}^n$ is the vector of actual states; $\hat{x} \in \mathbb{R}^n$ is the vector of estimated states; $y \in \mathbb{R}^p$ is the output vector; $u \in \mathbb{R}^m$ is the vector of control inputs; $\hat{A} \in \mathbb{R}^{n\times n}$ is the state matrix of the embedded model; $\hat{B} \in \mathbb{R}^{n\times m}$ is the control matrix of the embedded model; $\theta \in \mathbb{R}^{m\times m}$ is the actual fault parameter matrix; and $\hat{\theta} \in \mathbb{R}^{m\times m}$ is the estimated fault parameter matrix. Note that in this work, $\hat{A} = A$, $\hat{B} = B$, and $C = I$, though this is not generally true.

θ and $\hat{\theta}$ are square diagonal matrices whose entries are the multiplicative actuator faults for each of the control inputs. To determine $\hat{\theta}$, an estimate of the fault $\bar{\theta}$ must first be made, which is calculated through moving-horizon estimation (MHE) over a horizon of size N and sampling period Δ with the cost function,

$$J\left(\zeta_j, \bar{\theta}\right) = \sum_{i=j-N}^{j-1} \left(\left\| W\left(y_{i+1} - \bar{y}\left(\zeta_j, \bar{\theta}\right)_{i+1}\right)\right\|_2^2 \right), \tag{16}$$

where

$$\zeta_j = (x_{j-q}, u_{j-q}, \hat{\theta}_{j-q}, \Delta),\ q \in \{0, 1, 2, \ldots, N\}, \tag{17}$$

W is a weighting matrix, and $\|\cdot\|_2$ is the L2-norm. In application, at each sampling period i, an embedded model is initialized with $\bar{x}_{i-N} = x_{i-N}$ and, using the historical control inputs $u_{i-N\to i}$, the model is simulated from time index $i-N \to i$ using $\bar{\theta}$; figure (2) is provided for context.

The final estimation of the fault, θ_j, is the one that minimizes the cost function:

$$\theta_j = \underset{\bar{\theta}}{\operatorname{argmin}} \left\{ J\left(\zeta_j, \bar{\theta}\right) \right\}, \tag{18}$$

Once θ_j has been calculated, it can be used in determining a modified fault parameter $\hat{\theta}_j$ that may be able to improve the performance of the controller.

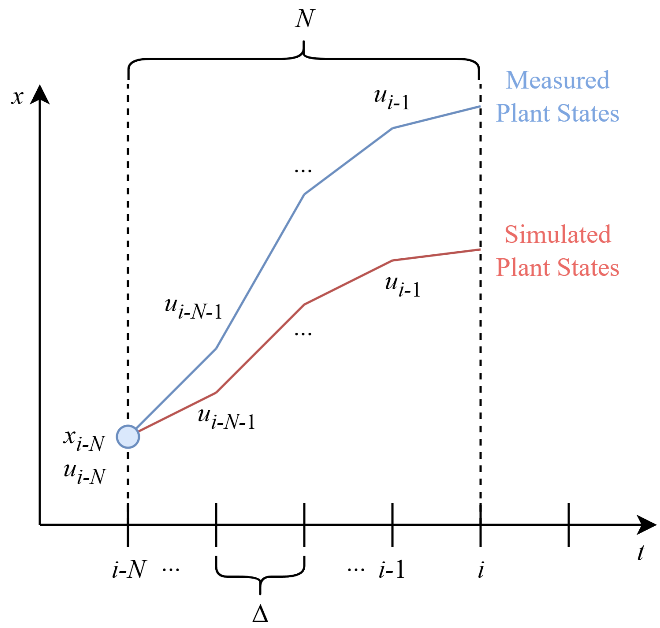


Fig. 2: Conceptual diagram of moving-horizon estimation. Our objective is to choose $\bar{\theta}$ such that the error between the measured plant states and the simulated plant states is minimized.

C. Fault Accommodation

The fault accommodation strategy involves choosing $\hat{\theta}$ at time index j such that,

$$\hat{\theta}_j = \underset{\hat{\theta}}{\operatorname{argmin}} \left\{ \|X\|_{\mathcal{H}_2} \right\}, \tag{19}$$

where $\|\cdot\|_{\mathcal{H}_2}$ is the $\mathcal{H}_2$-norm and X is the impulse response of the system over a horizon N_A in the time domain. Note that the choice of N_A is not dependent on N. Conceptually, the $\mathcal{H}_2$-norm is a measure of how quickly a system is able to recover from an impulse disturbance, where minimization of the $\mathcal{H}_2$-norm yields a system with a quicker disturbance response and shorter settling time—a high-performing system is characterized by a small $\mathcal{H}_2$-norm, and vice versa. The $\mathcal{H}_2$-norm also implicitly describes the stability of a system, where the solution to the Lyapunov equation is often required to calculate the norm [10]; this can be intuitively understood as a high-stability system has a small $\mathcal{H}_2$-norm, and vice versa.

Though the $\mathcal{H}_2$-norm is normally calculated explicitly, due to the implicit nature of MPC, we must numerically approximate it through simulation. To our benefit, $\|X\|_{\mathcal{H}_2}$ for discrete systems reduces to the Frobenius norm, $\|X\|_F$ ([10], [11]), making calculation straightforward. As the values of $\|X\|_F$ don't follow an analytical form, the approximate value of $\hat{\theta}_j$ is,

$$
\begin{aligned}
\hat{\theta}_j \approx \ & \underset{\hat{\theta}_i}{\operatorname{argmin}} \left\{ \|X\|_F(\hat{\theta}_i) \right\} \\
\text{s.t.} \quad & \|X\|_F(\hat{\theta}_i) - \|X\|_F(\hat{\theta}_{i-1}) \leq 0 \\
& \|X\|_F(\hat{\theta}_{i+1}) - \|X\|_F(\hat{\theta}_i) \geq 0
\end{aligned}
\tag{20}
$$

which estimates what value of $\hat{\theta}_j$ results in $\partial_{\hat{\theta}_j} \|X\|_F(\hat{\theta}_j) = 0$; the finer the gradations between $\hat{\theta}_i$ and $\hat{\theta}_{i+1}$ the more accurate the result, though of course incurring a higher computation cost.

Once $\hat{\theta}_j$ has been determined, the MPC algorithm can be updated using equation (14) and substituting $\hat{\theta} = \hat{\theta}_j$. An algorithm for this FTC estimation and accommodation strategy is expressed in algorithm (1), where the inputs are the history of measured states and control inputs from time index $i - N \rightarrow i$ ($X_{i-N\rightarrow i}$ and $U_{i-N\rightarrow i}$, respectively), the step size between $\hat{\theta}_i$ and $\hat{\theta}_{i+1}$ ($\Delta\hat{\theta}$), and the maximum number of steps to calculate $\hat{\theta}$ ($N_{\hat{\theta}}$) along with θ- and $\hat{\theta}$-estimation horizons (N, N_A, respectively); the outputs are θ_j and $\hat{\theta}_j$.

Note that the fault estimation and fault accommodation algorithms are not required to run at all times, and can be initiated based on event-triggered conditions. Also note that the fault accommodation strategy is not limited to calculating $\hat{\theta}_j$, and this same technique can be extended to calculate other parameters as minimizers of the $\mathcal{H}_2$-norm.

D. Fault-aware MPC cost function

The MPC cost function needs to account for the type of fault being modeled, where the constraints reflect the apparent loss/gain of actuator sensitivity to control inputs. The cost function maintains the usual form,

$$
J\left(\hat{x}, \hat{\theta}_j, u\right) = \hat{x}_{N_M}^\top P \hat{x}_{N_M} + \sum_{i=0}^{N_M-1} \left(\hat{x}_i^\top Q \hat{x}_i + u_i^\top R u_i\right), \tag{21}
$$

where N_M is the MPC horizon. We aim to find the minimum cost J^* through,

$$
J^*\left(\hat{\theta}_j\right) = \min_u \left\{ J\left(\hat{x}, \hat{\theta}_j, u\right) \right\}, \tag{22}
$$

subject to the constraints,

$$
\begin{gathered}
\hat{x}_{i+1} = \hat{A}\hat{x}_i + \hat{B}\hat{\theta}_j u_i \\
\hat{x}_{min} \leq \hat{x} \leq \hat{x}_{max} \\
u_{min} \leq u_i \leq u_{max},
\end{gathered}
$$

where the first constraint takes into account $\hat{\theta}_j$ in the state evolution. Of course, when there are no faults, $\hat{\theta}_j =$

Algorithm 1 Fault estimation and accommodation

Input: $X_{i-N\rightarrow i}$, $U_{i-N\rightarrow i}$, $\Delta\hat{\theta}$, $N_{\hat{\theta}}$, N, N_A
Output: θ_j, $\hat{\theta}_j$

```
    Fault Estimation
    Initialize: x̄_0 = X_{i-N}, θ̄ = I, θ_bool = False
    while θ_bool = False do
        for k = 0 to N − 1 do
            Simulate x̄_{k+1} = Â x̄_k + B̂ θ̄ U_{i−N−k}, store x̄_{0→N}
        end for
        if ‖X_{i−N→i} − x̄_{0→N}‖_F is not a minimum then
            Calculate θ̄ using nonlinear least-squares using the previous θ̄ to warm start
        else
            θ_j ← θ̄
            θ_bool ← True
        end if
    end while
    return θ_j
    Fault Accommodation
    Initialize: x̄_0 = X_{i−N}, x̂_0 = X_{i−N}, θ̂_n = I, θ̂_bool = False
    while θ̂_bool = False do
        for n = 0 to N_θ̂ − 1 do
            for l = 0 to N_A − 1 do
                Simulate x̄_{l+1} = Â x̄_l + B̂ θ_j u_l using MPC with embedded model x̂_{l+1} = Â x̂_l + B̂ θ̂_n u_l, store x̄_{0→N_A}
            end for
            Calculate F_n = ‖x̄_{0→N_A}‖_F, store F_n
            θ̂_n ← θ̂_n + Δθ̂
        end for
        if ∃θ̂_i s.t. equation (20) is satisfied then
            θ̂_j ← θ̂_i
            θ̂_bool ← True
        else
            θ̂_j ← θ_j
            θ̂_bool ← True
        end if
    end while
    return θ̂_j
```

I and the model states evolve as expected. However, as we are modeling the fault as affecting the strength of the actuator's effective response to a control input while still being bounded by the same minimum and maximum limits, the controller must be made aware of the scaling factor of the fault in the state-space model so as to not become saturated. The block diagram in figure (3) illustrates where the fault is modeled to occur.

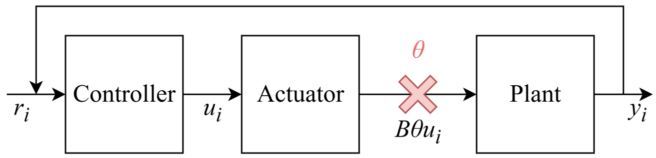


Fig. 3: The fault in the system occurs between the actuator and the plant. In other words, the actuator is healthy, however the link between the actuator and plant is faulty.

IV. Results and Discussion

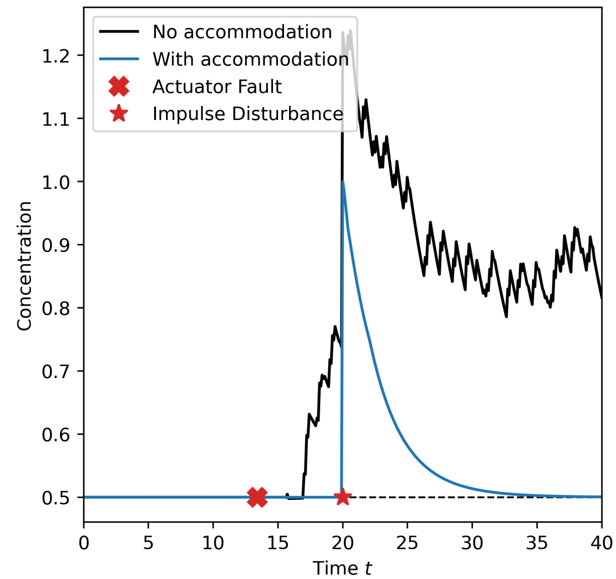
Unless otherwise stated, results were obtained by setting $\theta = I_\theta$, a square diagonal matrix whose entries are all the same constant θ. Q and R were chosen to be identity matrices, $Q = I_n$ and $R = I_m$. The sampling time period used is 0.1 seconds with a total time duration of 40 seconds. The MPC prediction horizon N_M is 10 steps. For sections (IV-A)-(IV-B), a fault estimation horizon of 10 steps was used. In section (IV-C), the horizon sensitivity study was swept over $N \in [2, 40]$. All MPC problems were solved using CVXPY, sequentially OSQP, CLARABEL, SCS, and then ECOS.

The state constraints were set as $C_i \in [0, 0.05]$ mol/m^3 for all species i. The reactor temperatures were set as $T_{react} \in [250, 600]$ K, and the jacket temperatures as $T_{cool} \in [200, 600]$ K. The input constraints for all flow rates were set as $u \in [0, u_{max}]$, where $u_{max} = 5$ m^3/s unless stated otherwise, such as section (IV-D). Similarly, the terminal cost was set as $P_N = Q$ for all simulations, except for section (IV-E), where $P_N = 0$. For the fault and disturbance simulations in sections (IV-A)-(IV-B), the actuator fault is implemented at $t = T/3$ and the impulse disturbance at $t = T/2$. Non-Gaussian measurement noise in sections (IV-A) and (IV-C) were modeled as $w(t) \in [-0.01, 0.01]$ which was added to the state measurements used in the fault estimation.

A. System response to fault and impulse disturbance

A model following the ODEs in equations (1)-(12) was developed. One instance was modeled using no fault estimation or accommodation, and another instance was modeled using the estimation and accommodation strategy outlined in algorithm (1). The results are observed in figure (4).

We see a marked improvement going from the system without accommodation (WOA) to the system with accommodation (WA). Whereas the WOA system experiences instability prior to the impulse disturbance, the WA system is able to adjust its control input to negate any instability. Once the impulse disturbance occurs, the WOA system is unable to drive the states to their equilibrium points, and the WA system drives them to their equilibrium points without any apparent issues.



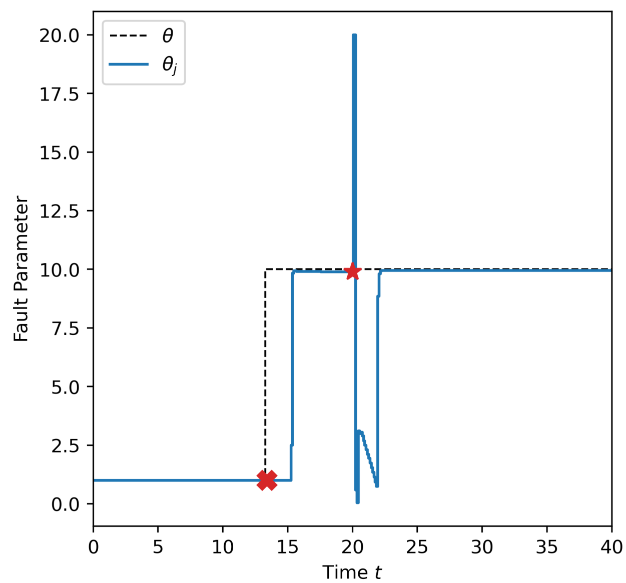


Fig. 4: (Top) Dynamic response of C_{A1} while the CSTR system undergoes an actuator fault ($\theta = 10$) with impulse disturbance $C_{A1} + 0.5$ mol/L without accommodation (black) and with accommodation (blue). (Bottom) Estimated fault parameter; fault estimation horizon $N = 10$.

We also observe the fault estimation behavior after a fault and after an impulse disturbance. Soon after the fault occurs, the fault estimation algorithm is able to deduce the severity of the fault, presumably through small deviations in the floating point values of the states, and obtain an estimated fault parameter close to the actual fault. After an impulse disturbance, the fault estimation must slowly "catch up" to the disturbance, and only once the entire estimation horizon is filled with faulted data is the estimation algorithm able to accurately deduce the fault severity.

Allen and El-Farra's work only investigated this fault estimation and accommodation method on deterministic systems with no uncertainties or measurement noise. However, MHE is known to be quite robust in terms of state estimation in stochastic processes. To probe the robustness of the fault estimation/accommodation method, non-Gaussian stochastic noise was added to the measurements used in estimating θ_j. The results of this simulation are observed in figure (5).

We see that the method is indeed able to estimate the magnitude of the fault with high accuracy when applied to a stochastic process. Further inspection reveals that the stochastic process is able to more quickly estimate the fault, as the noise provides deviations that the controller can use to estimate the fault before floating point errors/disturbances cause the deviation. MHE is thus, as expected, suitable to estimate faults for systems with nonlinearities and non-Gaussian uncertainties, and may even provide better results than deterministic processes when evaluated on certain grounds.

B. Estimation of multi-valued faults

The fault estimation and accommodation method is capable of handling several faults, each with a different magnitude. In figure (6) we observe the method correctly estimating the magnitudes of several faults and accommodating them such that the states stabilize and are driven to their equilibrium points; the impulse disturbance introduced was a spike in temperature in both CSTRs at the same time. Fault estimation and accommodation of several actuators can thus happen reliably and independently of each other.

C. $\mathcal{H}_2$-norm dependence on estimation horizon size N

The size of the fault estimation horizon N is a critical parameter in determining the performance of the closed-loop system. An optimal N would respond to system disturbances quickly (favoring a smaller N), though also provide the most accurate estimate of the fault magnitude (favoring a larger N). To determine the optimal N for

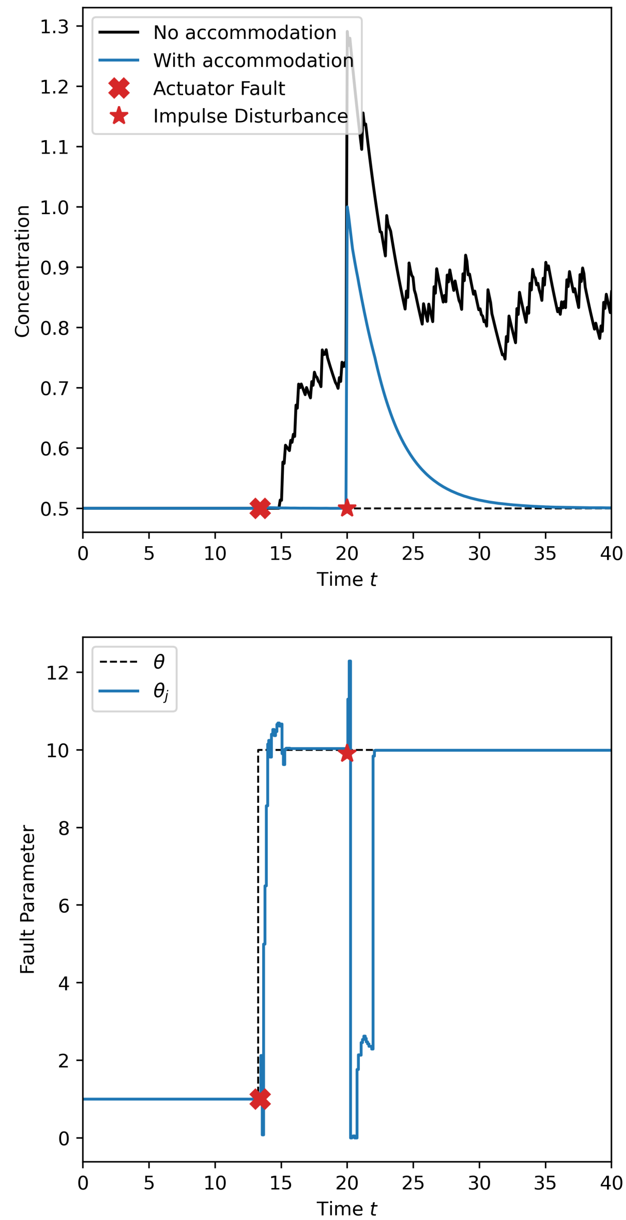


Fig. 5: Dynamic response of C_{A1} with measurement noise, $w(t) \in [-0.01, 0.01]$. The fault estimation method is able to deal with systems where measurement noise is present.

this particular system, a range of N was used to calculate the $\mathcal{H}_2$-norm given identical impulse disturbances across simulations, which we observe in figure (7a).

We can see that, based on the scale of the vertical axis, the affect of horizon size is relatively small for this particular system, though if we truly needed to choose the best option, it seems that $N = 23$ yields the highest-performing system. These simulations were

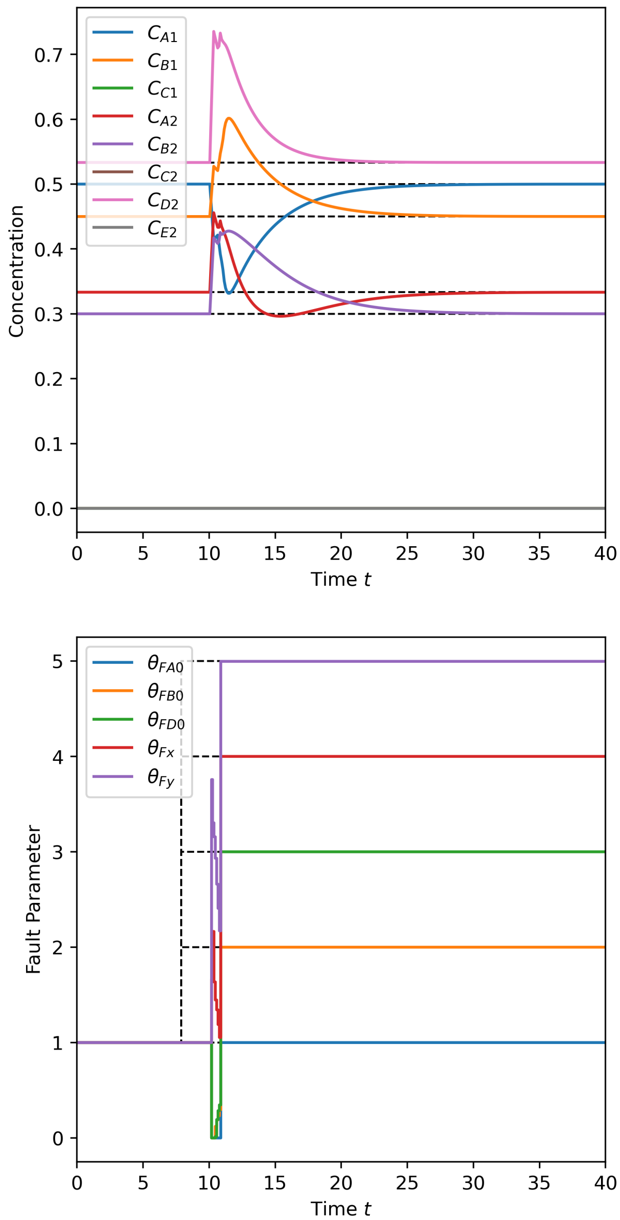
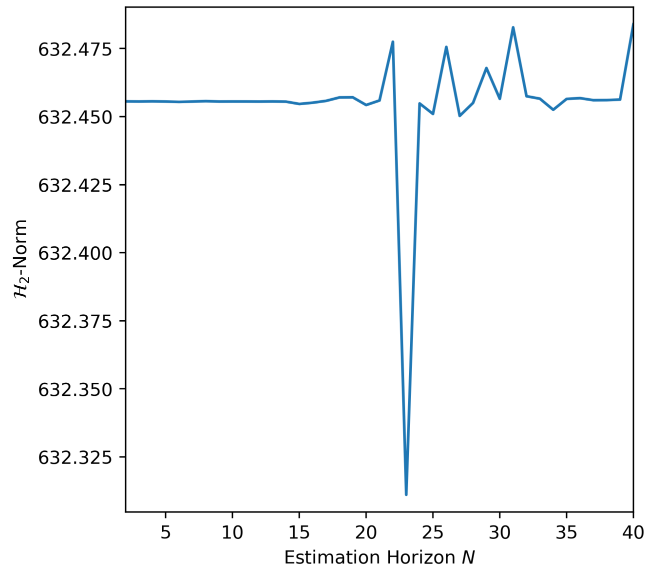
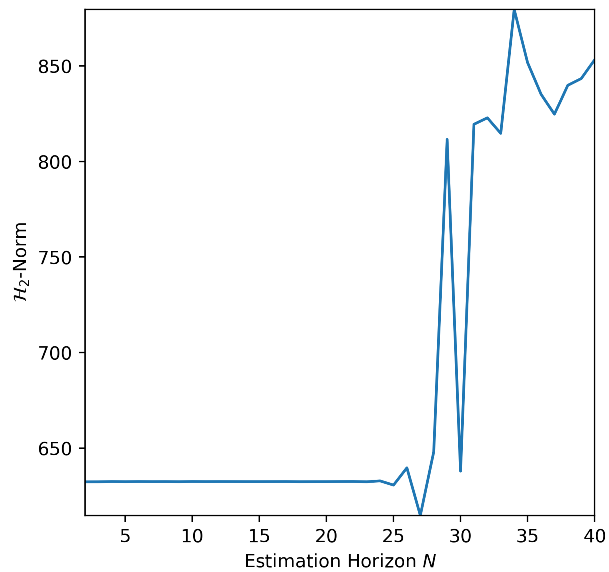


Fig. 6: Fault estimation and accommodation when the system undergoes faults of different magnitudes: $\theta = [1, 2, 3, 4, 5]$.



(a) Deterministic



(b) Stochastic

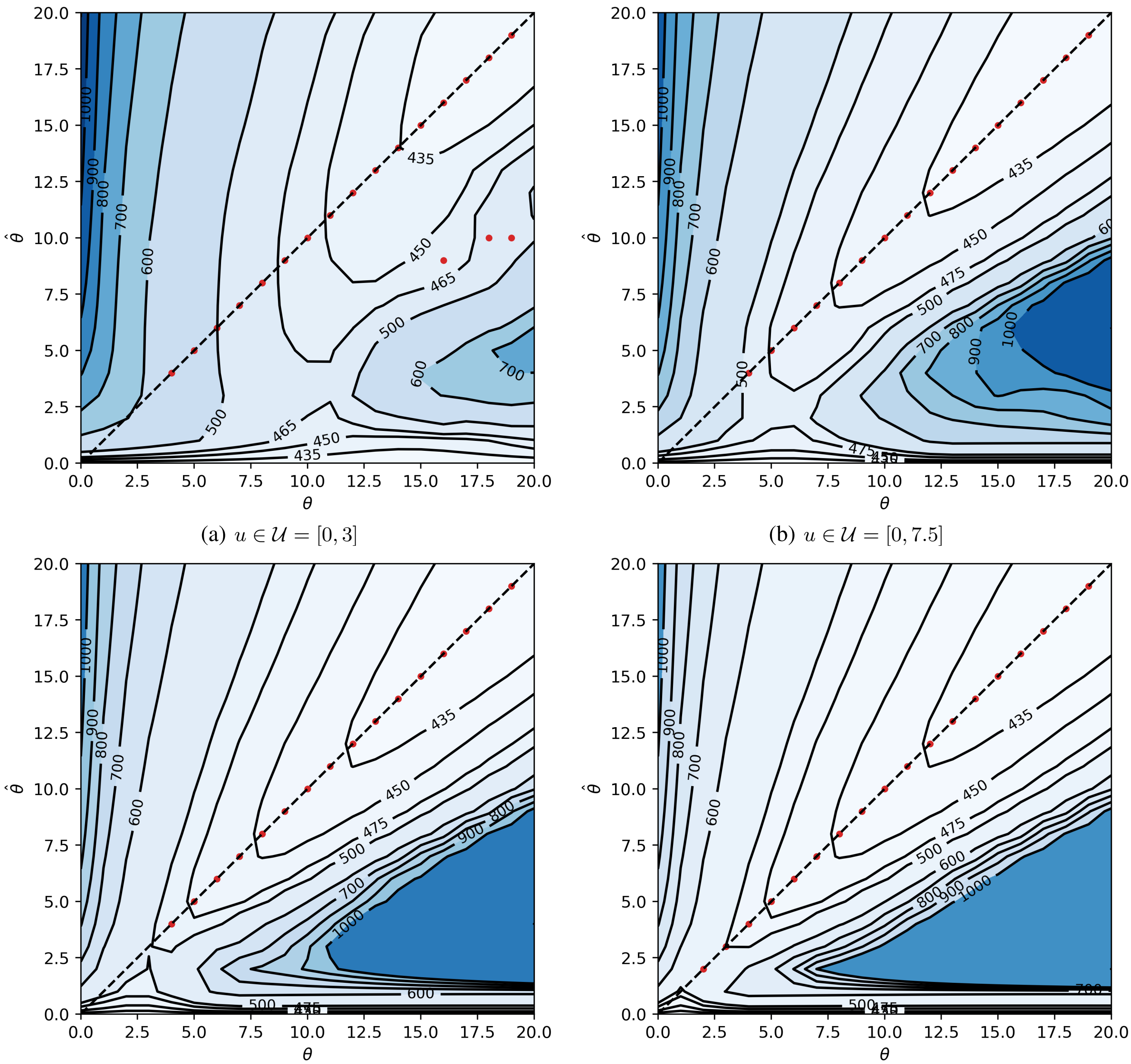
Fig. 7: The $\mathcal{H}_2$-norm sees a relatively weak dependence on estimation horizon size N for this particular system in the deterministic simulation. However, adding noise we find that $N \leq 25$ is favorable.

done entirely deterministically, where the horizon size contributes a relatively smaller role for LTI systems as the plant evolves exactly as the model predicts.

We again applied the fault accommodation method to the model with measurement noise, seen in figure (7b), where we find that a horizon of up to $N \approx 25$ yields a system with comparable performance to the deterministic system. Once the horizon gets larger, however, the estimation and accommodation of the fault lags behind, resulting in worse overall performance despite yielding a more accurate fault estimation.

D. $\mathcal{H}_2$-norm dependence on admissible inputs $\mathcal{U}$

The main benefit of MPC is its ability to account for state and input constraints; it must then come as no surprise that the set of admissible inputs $\mathcal{U}$ is a key parameter that affects how quickly a system can respond to a disturbance, thus affecting the $\mathcal{H}_2$-norm. Figure (8)



(a) $u \in \mathcal{U} = [0, 3]$

(b) $u \in \mathcal{U} = [0, 7.5]$

(c) $u \in \mathcal{U} = [0, 15]$

(d) $u \in \mathcal{U} = [0, \infty)$

Fig. 8: Contour plots of the $\mathcal{H}_2$-norm with different input constraints. Red areas indicate a high $\mathcal{H}_2$-norm, blue areas indicate a low $\mathcal{H}_2$-norm. The dashed 45-degree line indicates the accommodation strategy where $\hat{\theta}_j = \theta_j$. The figures also feature red dots that indicate where equation (20) finds local minima at a given θ_j.

shows the effects of changing the maximum control input in $\mathcal{U}$, where increasing u_{max} yields various contours that show the areas of low- and high-performing systems.

The contours of the $\mathcal{H}_2$-norm surface show that the most high-performing configuration is the one where the controller believes a "desensitizing" fault ($\hat{\theta}_j \leq 1$ for all θ_j) or "super-sensitizing" fault ($\theta_j \geq 15$ and $\hat{\theta}_j \geq 15$) has occurred. In the case of a desensitizing fault, the change in effective actuator sensitivity must be counteracted by an increase in the calculated control effort, resulting in an actual response that is quicker. In the case of a super-sensitizing fault, the system itself is responding more strongly to the control input, and all the controller needs to do is set $\hat{\theta}_j = \theta_j$ to take advantage of the increased sensitivity. The figures also feature red dots where equation (20) finds a local minimum at a given θ_j, where we observe that the $\hat{\theta}_j = \theta_j$ line most often yields a local minimum. These local minima are present in all subfigures, which suggests that the $\hat{\theta}_j = \theta_j$ line is among the highest-performing $\hat{\theta}_j$-θ_j configurations available.

In figure (9) we can see C_{A1} profiles as $\hat{\theta}_j$ is adjusted after a fault of $\theta = 15$ has occurred—it must be noted that the $\mathcal{H}_2$-norm calculates the norm based on all the states, meaning that though the $\hat{\theta}_j = 12.5$ profile reaches the equilibrium state quicker, the deviation of the rest of the states makes the norm larger. With this in mind, we see trends that match our expectations: $\hat{\theta}_j$-θ_j pairs that lie within specific regions of the $\mathcal{H}_2$-norm surface yield state trajectories that are unstable (very large norm), stable yet underdamped (large norm), or stable with no overshoot (small norm). We observe interesting behavior where $\hat{\theta}_j = 1$ will yield a stable closed-loop system when $\hat{\theta}_j = 2.5$ doesn't, which may mean there are $\hat{\theta}_j$:θ_j ratios that have inherent stability for a given system.

We observe the regions of high performance become more pronounced when u_{max} becomes larger. This is due to the controller being more free in its trajectory within input space, allowing it to provide larger control inputs to better handle large errors between the states and their equilibrium points.

We also see that the increasing u_{max} results in the $\hat{\theta}_j = \theta_j$ line to become increasingly favorable as it begins to occupy some of the most high-performing regions of the parameter space. An interpretation of this is that when input constraints become more loose, the state trajectory obtained from MPC is not only optimal in terms of state and input costs, but is also among the highest-performing trajectories in terms of the $\mathcal{H}_2$-norm. Interpreting this further, as this system was modeled as LTI and deterministic, the trajectory obtained through MPC with extremely loose constraints approaches that of discrete, finite-horizon LQR—by this logic, it seems that for control problems where state and input constraints are not active, LQR produces both the optimal and highest-performing state trajectory. This conclusion is corroborated by known equivalencies between LQR and $\mathcal{H}_2$-norm control [12].

There also exists a region at the bottom right of all the contours where the $\mathcal{H}_2$-norm is high, indicating that the closed-loop system is unstable. In these configurations, it is detrimental to have a controller provide inputs to the system and the system would benefit more from implementing $\hat{\theta}_j$-decision logic to actively avoid this region.

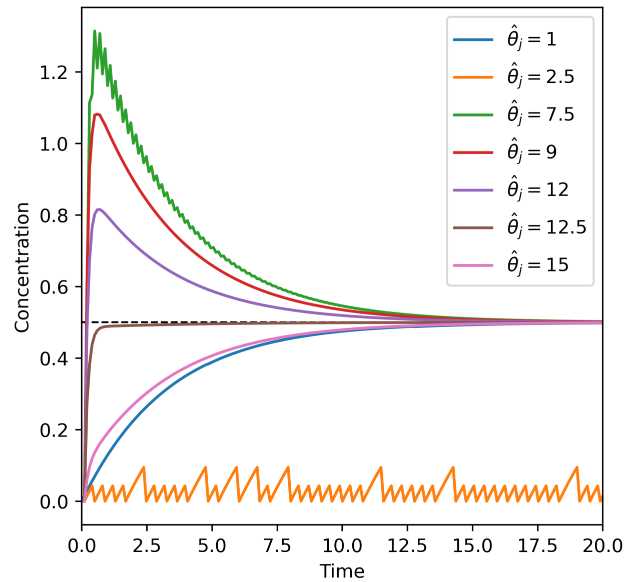


Fig. 9: C_{A1} profiles obtained from controllers using different $\hat{\theta}_j$ values.

E. Comparison between MPC and LQR

As mentioned in the introduction, MPC is often favored over other model-based control methods for its optimal constraint handling. Thus, we compared the dynamic response of a constrained system controlled via MPC and discrete LQR to quantify how much more optimal it is in terms of the quadratic cost function and $\mathcal{H}_2$-norm. The control methods are assumed to already be aware of the fault ($\theta = 10$) and have made the optimal accommodation decision ($\hat{\theta}_j = 10$). The LQR gain K was calculated by considering the state-space model with a linear control law,

$$x_{k+1} = A_d x_k + B_d u_k \tag{23}$$
$$u_k = -K x_k, \tag{24}$$

where $A_d = e^{A\Delta}$ and $B_d = A^{-1}(e^{A\Delta} - I)B\theta$, allowing the calculation of K to account for the fault. Figure (10) shows the results of the simulations using LQR and MPC responding to non-equilibrated initial conditions.

We observe the LQR response (black) settles slower than MPC, due to the controller attempting to drive the concentration negative without considering that the state is fundamentally bounded. MPC, by contrast, is able to settle much quicker as it takes the state constraints into account. This is true even with the MPC terminal cost $P_N = 0$ and horizon size set to the length of the simulation ($N_M = 200$). The resulting quadratic costs over the simulation period were $J_{LQR} = 4.5 \times 10^4$ and

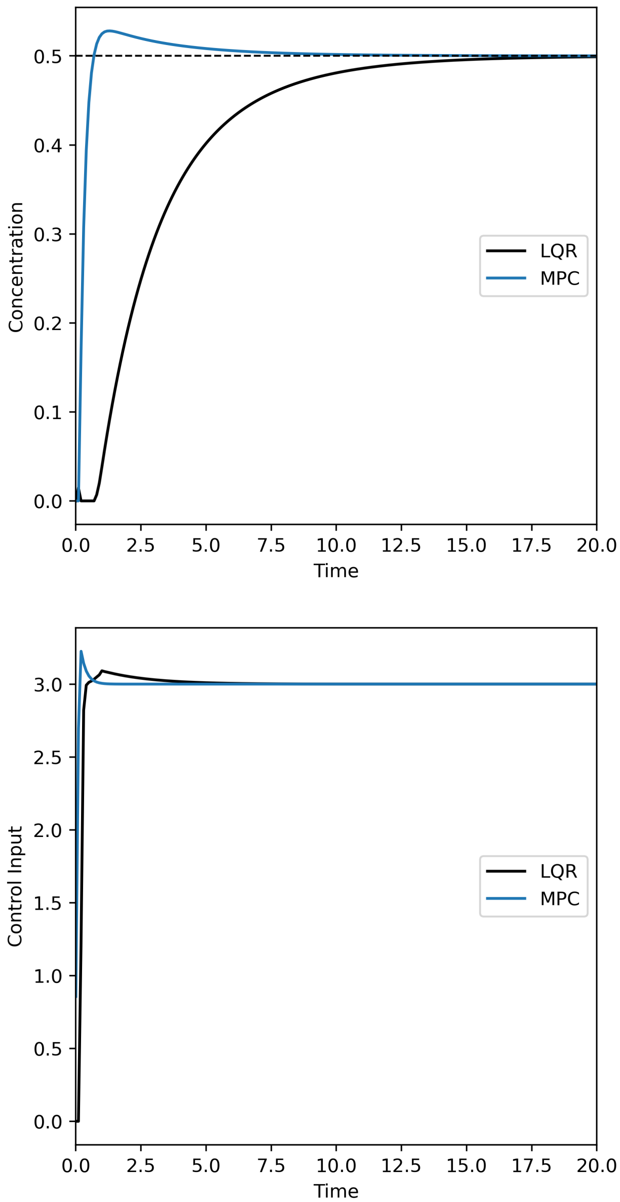


Fig. 10: Dynamic responses for a faulted system controlled through discrete LQR (black) and MPC (blue). MPC produces a state and input trajectory that results in both a smaller quadratic cost and smaller $\mathcal{H}_2$-norm.

$J_{MPC} = 1.1 \times 10^4$; the $\mathcal{H}_2$-norms over the simulation period were $\mathcal{H}_{2,LQR} = 734$ and $\mathcal{H}_{2,MPC} = 438$. This results in $J_{LQR} > J_{MPC}$ and $\mathcal{H}_{2,LQR} > \mathcal{H}_{2,MPC}$, which agree with our expectation.

These results further validate the argument that MPC is preferable over unconstrained control methods, especially for systems where performance is of importance.

V. Directions for Future Work

There are many branches of FTC that can be explored stemming from the idea of $\mathcal{H}_2$-norm minimization to obtain an updated fault parameter. As this work explored the application of this method to an MPC system, future work can expand upon the formulation of the model to account for multiplicative and additive actuator faults, along with plant-model mismatches. Such models would be among the most accurate in terms of describing the effects of possible faults. An example state-space model could be

$$x_{k+1} = Ax_k + B(\theta_m u_k + \theta_a) \tag{25}$$

$$\hat{x}_{k+1} = \hat{A}\hat{x} + \hat{B}(\hat{\theta}_m u_k + \hat{\theta}_a) + L(y_k - \hat{y}_k) \tag{26}$$

$$\hat{y}_k = C\hat{x} \tag{27}$$

where $\hat{A} \neq A$ and $\hat{B} \neq B$. This generalization can be taken a step further where the system is modeled as stochastic. In this formulation, one must develop and algorithm that accurately estimates several faults of varying magnitude and behavior. The use of observers may also prove challenging, and it is possible that its inclusion makes the MPC algorithm intractable.

Alternatively, one could choose to model different types of fault behaviors. This work focuses on faults that affect the link between the actuator and the plant, however there are several types of faults to consider, such as faults that affect the controller-actuator link (similar to faults found in networked control systems undergoing false data injections via cyberattack [13]). However, with MPC's ability to consider state and input constraints, one could choose to apply a multiplicative or additive fault parameter to any aspect of the MPC formulation. An example would be to add it to the input constraint, which represents the reference signal to the controller being faulty:

$$\begin{aligned} J^*\left(\hat{\theta}_j\right) &= \min_u \left\{ J\left(\hat{x}, \hat{\theta}_j, u\right) \right\} \\ \text{s.t} \quad & \hat{x}_{i+1} = \hat{A}\hat{x}_i + \hat{B}u_i \\ & \hat{x}_{min} \leq \hat{x} \leq \hat{x}_{max} \\ & u_{min} \leq \hat{\theta}_j u_i \leq u_{max}. \end{aligned} \tag{28}$$

Another direction that research can take is investigation into multiplicative or additive nonlinear fault parameters, such as those that follow a logistic/sigmoidal curve. This route will require nonlinear MPC (NMPC), which is able to better model real-world behaviors.

As we briefly studied the effect of applying the fault estimation/accommodation method to stochastic systems, one could also attempt analyzing the behavior of the system using robust or stochastic MPC.

A final suggestion is to find the extent to which the results found in Napasindayao's and Allen's work can be translated from explicit control laws to implicit MPC, where the analytical strength of explicit control laws may be paired with the optimal constraint handling of MPC. This route will require extensive work to bridge the gap between the explicit and implicit worlds of system dynamics and control.

VI. Conclusion

Our exploration on applying an $\mathcal{H}_2$-norm minimization approach to a system controlled by MPC found that fault accommodation is often the highest-performing when the accommodated fault parameter is equal to the estimated fault, indicating that minimal accommodation-related calculations are required once the fault is estimated. The fault estimation and accommodation method was also shown to work well with multiple faults, and that the method using MPC produces a state and input trajectory that results in a smaller quadratic cost and $\mathcal{H}_2$-norm compared to LQR. We also explore the effects of changing the size of the set of input constraints and how it affects the shape of the $\mathcal{H}_2$-norm surface, where we see a clear dependence of performance on maximum input constraints. The effects of modeling a deterministic versus stochastic process were also briefly explored, though more work will need to be done to explore the resulting behavior more thoroughly. There are many possible lines of research that branch from this exploratory study, where functional analysis and nonlinear MPC will likely be needed to investigate the behavior of systems where faults are modeled more thoroughly.

VII. Contributions

Member	Main Contributions
Frank	Problem conceptualization, methodology, system modeling and simulation, writing, reviewing and editing manuscripts and presentation
Jaden	System modeling and simulation, writing, reviewing and editing manuscripts and presentation, presentation slides

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Appendix A
Table of Variables

TABLE I: Variables, their description, and values.

Variable	Description	Value
C_{A1}^{init}	Initial concentration of A, CSTR 1	0 mol/m^3
C_{B1}^{init}	Initial concentration of B, CSTR 1	0.03 mol/m^3
C_{C1}^{init}	Initial concentration of C, CSTR 1	0 mol/m^3
C_{A2}^{init}	Initial concentration of A, CSTR 2	0 mol/m^3

C_{B2}^{init}	Initial concentration of B, CSTR 2	0 mol/m^3
C_{C2}^{init}	Initial concentration of C, CSTR 2	0.01 mol/m^3
C_{D2}^{init}	Initial concentration of D, CSTR 2	0.015 mol/m^3
C_{E2}^{init}	Initial concentration of E, CSTR 2	0.01 mol/m^3
C_{A0}	Inlet concentration of A	1 mol/m^3
C_{B0}	Inlet concentration of B	0.001 mol/m^3
C_{D0}	Inlet concentration of D	0.002 mol/m^3
T_{x0}	Inlet temperature of x	298 Kelvin
T_{y0}	Inlet temperature of y	250 Kelvin
F_{A0}	Initial volumetric flow rate of A	0 m^3/s
F_{B0}	Initial volumetric flow rate of B	0 m^3/s
F_{D0}	Initial volumetric flow rate of D	0 m^3/s
F_{x0}	Initial volumetric flow rate of x	0 m^3/s
F_{y0}	Initial volumetric flow rate of y	0 m^3/s
F_A^{sp}	Equi. volumetric flow rate of A	3 m^3/s
F_B^{sp}	Equi. volumetric flow rate of B	3 m^3/s
F_D^{sp}	Equi. volumetric flow rate of D	3 m^3/s
F_x^{sp}	Equi. volumetric flow rate of x	1 m^3/s
F_y^{sp}	Equi. volumetric flow rate of y	1 m^3/s
C_{A1}^{sp}	Equi. concentration of A, CSTR 1	0.5 m^3/s
C_{B1}^{sp}	Equi. concentration of B, CSTR 1	0.45 m^3/s
C_{C1}^{sp}	Equi. concentration of C, CSTR 1	2.3e-6 mol/m^3
C_{A2}^{sp}	Equi. concentration of A, CSTR 2	0.33 mol/m^3
C_{B2}^{sp}	Equi. concentration of B, CSTR 2	0.30 mol/m^3
C_{C2}^{sp}	Equi. concentration of C, CSTR 2	3.4e-5 mol/m^3
C_{D2}^{sp}	Equi. concentration of D, CSTR 2	0.53 mol/m^3
C_{E2}^{sp}	Equi. concentration of E, CSTR 2	0.0 mol/m^3
T_1^{sp}	Equi. temperature of CSTR 1	241.4 Kelvin
T_2^{sp}	Equi. temperature of CSTR 2	285.7 Kelvin
T_x^{sp}	Equi. temperature of x	287.1 Kelvin
T_y^{sp}	Equi. temperature of y	256.9 Kelvin
V_1	Volume of CSTR 1	20 m^3
V_2	Volume of CSTR 2	20 m^3
V_x	Volume of heat jacket for CSTR 1	1 m^3
V_y	Volume of heat jacket for CSTR 2	1 m^3
A_1	Frequency factor, R_1	1e+5 m^{3}mol^{-1}s^{-1}
A_2	Frequency factor, R_2	1e+8 m^{3}mol^{-1}s^{-1}
E_1	Activation energy, R_1	5e+4 J/mol
E_2	Activation energy, R_2	1e+4 J/mol
R_g	Ideal gas constant	8.314 J mol^{-1}K^{-1}
ΔH_1	Enthalpy of reaction, R_1	1e+5 J/mol
ΔH_2	Enthalpy of reaction, R_2	5e+5 J/mol
U_x	Overall heat transfer coeff. for CSTR 1	100 W m^{-2}K^{-1}
U_y	Overall heat transfer coeff. for CSTR 2	100 W m^{-2}K^{-1}
A_x	Heat transfer area for CSTR 1	10 m^2
A_y	Heat transfer area for CSTR 2	10 m^2
C_{pr}	Heat capacity for reactive mixture	4.184 J/K
C_{pc}	Heat capacity for cooling water	4.184 J/K
ρ_r	Mass density for reactive mixture	1000 kg/m^3
ρ_c	Mass density for cooling water	1000 kg/m^3